MATERIAL MODELING 9

CS 184: FOUNDATIONS OF COMPUTER GRAPHICS

1 BRDF Physical Plausibility

A physically plausible BRDF must satisfy certain properties. Consider the following BRDF:

$$f_r(\omega_i, \omega_o) = \frac{k\,(\omega_i \cdot \omega_o)^n}{(\omega_i \cdot \mathbf{n})(\omega_o \cdot \mathbf{n})}$$

where k and n are constants, ω_i is the incoming direction, ω_o is the outgoing direction, and $\mathbf{n}$ is the surface normal.

1. Check if this BRDF satisfies positivity.

   Solution: Positivity requires that $f_r(\omega_i, \omega_o) \geq 0$ for all valid ω_i and ω_o.

   For physically valid directions (above the surface), we have $(\omega_i \cdot \mathbf{n}) \geq 0$ and $(\omega_o \cdot \mathbf{n}) \geq 0$.

   If $k \geq 0$ and n is even, then $(\omega_i \cdot \omega_o)^n \geq 0$ for all directions. If n is odd, then $(\omega_i \cdot \omega_o)^n$ can be negative when $(\omega_i \cdot \omega_o) < 0$.

   Therefore, the BRDF satisfies positivity if $k \geq 0$ and n is even.

2. Check if this BRDF satisfies Helmholtz reciprocity (symmetry).

   Solution: Helmholtz reciprocity requires that $f_r(\omega_i, \omega_o) = f_r(\omega_o, \omega_i)$. Let's check:

   $$f_r(\omega_i, \omega_o) = \frac{k\,(\omega_i \cdot \omega_o)^n}{(\omega_i \cdot \mathbf{n})(\omega_o \cdot \mathbf{n})}$$

   Swapping ω_i and ω_o:

   $$f_r(\omega_o, \omega_i) = \frac{k\,(\omega_o \cdot \omega_i)^n}{(\omega_o \cdot \mathbf{n})(\omega_i \cdot \mathbf{n})}$$

   Since the dot product is commutative, $(\omega_i \cdot \omega_o) = (\omega_o \cdot \omega_i)$. Therefore:

   $$f_r(\omega_o, \omega_i) = \frac{k\,(\omega_i \cdot \omega_o)^n}{(\omega_o \cdot \mathbf{n})(\omega_i \cdot \mathbf{n})} = f_r(\omega_i, \omega_o)$$

   So the BRDF satisfies Helmholtz reciprocity.

3. Find the constraints on k and n that would make this BRDF conserve energy.

Solution: Energy conservation requires that for any incoming direction ω_i, the total energy reflected in all directions cannot exceed the energy received:

$$\int_\Omega f_r(\omega_i, \omega_o)\,(\omega_o \cdot \mathbf{n})\,\mathrm{d}\omega_o \leq 1$$

Substituting our BRDF:

$$\int_\Omega \frac{k\,(\omega_i \cdot \omega_o)^n}{(\omega_i \cdot \mathbf{n})(\omega_o \cdot \mathbf{n})}\,(\omega_o \cdot \mathbf{n})\,\mathrm{d}\omega_o = \frac{k}{(\omega_i \cdot \mathbf{n})}\int_\Omega (\omega_i \cdot \omega_o)^n\,\mathrm{d}\omega_o$$

This integral depends on the value of n. For energy conservation, we need:

$$\frac{k}{(\omega_i \cdot \mathbf{n})}\int_\Omega (\omega_i \cdot \omega_o)^n\,\mathrm{d}\omega_o \leq 1 \quad \text{for all } \omega_i$$

The worst case is when $\omega_i = \mathbf{n}$ (normal incidence), which maximizes the integral. In this case:

$$k\int_\Omega (\cos\theta_o)^n\,\mathrm{d}\omega_o \leq 1$$

The integral evaluates to $\frac{2\pi}{n+1}$ for even n. Therefore:

$$k \leq \frac{n+1}{2\pi}$$

This is the constraint on k for energy conservation, assuming n is even (which we need for positivity).

2 Cook-Torrance BRDF Calculation

This problem will walk you through a full shading calculation for a single point on a surface using the Cook-Torrance microfacet model.

Scenario:

- A point on a surface has normal $\mathbf{n} = (0, 0, 1)$.
- The viewer is in direction $\omega_o = \left(\frac{\sqrt{3}}{2}, 0, \frac{1}{2}\right)$.
- A single point light provides radiance L_i from direction $\omega_i = \left(-\frac{\sqrt{3}}{2}, 0, \frac{1}{2}\right)$.
- Assume for this setup, the material properties evaluate to $D = 1/\pi$, $G = 1$, and $F = F_0$.

Your task is to calculate the final outgoing radiance, $L_o(\omega_o)$.

1. Calculate the halfway vector, h.

Solution:

$$h = \frac{\omega_i + \omega_o}{\|\omega_i + \omega_o\|} = \frac{\left(-\frac{\sqrt{3}}{2}, 0, \frac{1}{2}\right) + \left(\frac{\sqrt{3}}{2}, 0, \frac{1}{2}\right)}{\|\dots\|} = \frac{(0, 0, 1)}{\|(0, 0, 1)\|} = (0, 0, 1)$$

The halfway vector is aligned with the normal, representing a perfect specular reflection condition.

2. Calculate the dot products $(\mathbf{n} \cdot \omega_i)$ and $(\mathbf{n} \cdot \omega_o)$.

Solution:

$$(\mathbf{n} \cdot \omega_i) = (0, 0, 1) \cdot \left(-\frac{\sqrt{3}}{2}, 0, \frac{1}{2}\right) = \frac{1}{2}$$
$$(\mathbf{n} \cdot \omega_o) = (0, 0, 1) \cdot \left(\frac{\sqrt{3}}{2}, 0, \frac{1}{2}\right) = \frac{1}{2}$$

3. Assemble the full Cook-Torrance BRDF value, $f_r = \frac{D \cdot G \cdot F}{4(\mathbf{n} \cdot \omega_i)(\mathbf{n} \cdot \omega_o)}$.

Solution:

$$f_r = \frac{D \cdot G \cdot F}{4(\mathbf{n} \cdot \omega_i)(\mathbf{n} \cdot \omega_o)} = \frac{(1/\pi) \cdot 1 \cdot F_0}{4 \cdot (1/2) \cdot (1/2)} = \frac{F_0/\pi}{1} = \frac{F_0}{\pi}$$

4. Calculate the final outgoing radiance L_o using the reflection equation for a single point light: $L_o = f_r \cdot L_i \cdot (\mathbf{n} \cdot \omega_i)$.

Solution:

$$L_o = f_r \cdot L_i \cdot (\mathbf{n} \cdot \omega_i) = \frac{F_0}{\pi} \cdot L_i \cdot \frac{1}{2} = \frac{F_0 L_i}{2\pi}$$